ITCS285 Database Management Systems

Second Semester, 2025-2026

Lab #1

Type SQL queries to answer the following:

Q1: Create all tables with their related constraints.

```
SQL> CREATE TABLE Client (
  2     ClientID   NUMBER(5)    PRIMARY KEY,
  3     FirstName  VARCHAR2(50) NOT NULL,
  4     LastName   VARCHAR2(50) NOT NULL,
  5     Phone      VARCHAR2(20),
  6     Address    VARCHAR2(100),
  7     Profession VARCHAR2(50)
  8  );
Table created.

SQL> CREATE TABLE BANK(
  2  BankID NUMBER(5)PRIMARY KEY,
  3  BankName VARCHAR2(50) NOT NULL
  4  );
Table created.

SQL> CREATE TABLE Branch (
  2     BranchID      VARCHAR2(10)  PRIMARY KEY,
```

```
  3     BankID       NUMBER(5)     NOT NULL,
  4     BranchName    VARCHAR2(50),
  5     BranchAddress VARCHAR2(100),
  6     CONSTRAINT fk_branch_bank FOREIGN KEY (BankID) REFERENCES Bank(BankID)
  7  );
```

Table created.

```
SQL> CREATE TABLE Account (
  2     AccountID VARCHAR2(10)  PRIMARY KEY,
  3     BranchID  VARCHAR2(10)  NOT NULL,
  4     ClientID  NUMBER(5)     NOT NULL,
  5     Balance   NUMBER(10,2),
  6     CONSTRAINT fk_account_branch FOREIGN KEY (BranchID) REFERENCES Branch(BranchID),
  7     CONSTRAINT fk_account_client FOREIGN KEY (ClientID) REFERENCES Client(ClientID)
  8  );
```

Table created.

Q2: Insert data into the different tables

```
SQL> INSERT INTO Client VALUES (10000, 'Ahmad', 'Abd Allah', '2134657', 'Manama',
'Professor');
```

1 row created.

```
SQL> INSERT INTO Client VALUES (10001, 'Ghazi', 'Al-Amri', '2345676', 'Manama', 'Student');
```

1 row created.

```
SQL> INSERT INTO Client VALUES (10002, 'Mohamed', 'Al-Hamed', '5467698', 'Manama',
'Trader');

1 row created.


SQL> INSERT INTO Client VALUES (10003, 'Ali', 'Hussain', '6578790', 'Manama', 'Teller');

1 row created.


SQL> INSERT INTO Client VALUES (10004, 'Amer', 'Al-Salah', '6578798', 'Manama', 'Professor');

1 row created.


SQL> INSERT INTO Client VALUES (10005, 'Salah', 'Al-Kacem', '6579879', 'Manama', 'Engineer');

1 row created.


SQL> INSERT INTO Bank VALUES(30001, 'BBK');

1 row created.


SQL> INSERT INTO Bank VALUES (30002, 'NBB');

1 row created.


SQL> INSERT INTO Bank VALUES (30003, 'CITI');

1 row created.


SQL> INSERT INTO Branch VALUES ('A100', 30003, 'CITYCENTER', 'Manama');

1 row created.


SQL> INSERT INTO Branch VALUES ('A101', 30003, 'HAMAD TOWN', 'HAMALAH');
```

```
1 row created.

SQL> INSERT INTO Branch VALUES ('R100', 30001, 'ZALLAQ', 'ZALLAQ');
1 row created.

SQL> INSERT INTO Branch VALUES ('R101', 30001, 'SEEF', 'Manama');
1 row created.

SQL> INSERT INTO Account VALUES ('AC100', 'A100', 10000, 20000);
1 row created.

SQL> INSERT INTO Account VALUES ('AC101', 'A100', 10001, 120000);
1 row created.

SQL> INSERT INTO Account VALUES ('AC102', 'A101', 10002, 25300);
1 row created.

SQL> INSERT INTO Account VALUES ('AC103', 'R100', 10003, 26500);
1 row created.

SQL> INSERT INTO Account VALUES ('AC104', 'R100', 10004, 45600);
1 row created.

SQL> INSERT INTO Account VALUES ('AC105', 'R101', 10004, 56000);
1 row created.
```

```
SQL> COMMIT;

Commit complete.
```

Q3: Modify table Bank to add a new attribute BankCode of Type string with exactly 8 characters.

```
SQL> ALTER TABLE Bank ADD BankCode CHAR(8);

Table altered.
```

Q4: Find clients with first names starting with A and having at least 3 characters.

```
SQL> SELECT *
  2  FROM Client
  3  WHERE FirstName LIKE 'A%'
  4  AND LENGTH(FirstName) >= 3;


  CLIENTID FIRSTNAME
---------- --------------------------------------------------
LASTNAME                                           PHONE
-------------------------------------------------- --------------------
ADDRESS
--------------------------------------------------------------------------------
PROFESSION
--------------------------------------------------
     10000 Ahmad
Abd Allah                                          2134657
Manama
Professor
```

CLIENTID FIRSTNAME

---------- ----------

LASTNAME PHONE

---------- ----------

ADDRESS

PROFESSION

10003 Ali

Hussain 6578790

Manama

Teller

CLIENTID FIRSTNAME

---------- ----------

LASTNAME PHONE

---------- ----------

ADDRESS

PROFESSION

10004 Amer

Al-Salah 6578798

Manama

Professor